

Practical 1: VFS Architecture and File Operations

```
saireddy@DESKTOP-HKQRL9G: ~  
GNU nano 7.2  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <sys/stat.h>  
  
int main()  
{  
    int fd;  
    char buffer[100];  
  
    fd = open("test.txt", O_CREAT | O_RDWR, 0644);  
  
    if (fd < 0)  
    {  
        perror("open");  
        return 1;  
    }  
  
    write(fd, "Hello VFS in Linux", 18);  
  
    lseek(fd, 0, SEEK_SET);  
  
    int n = read(fd, buffer, 18);  
    buffer[n] = '\0';  
  
    printf("File Content: %s\n", buffer);  
  
    struct stat st;  
    fstat(fd, &st);  
  
    printf("File Size: %ld bytes\n", (long)st.st_size);  
    printf("Inode Number: %ld\n", (long)st.st_ino);  
  
    close(fd);  
  
    return 0;  
}
```

```
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical$ nano vfs.c  
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical$ gcc vfs.c -o vfs  
./vfs  
File Content: Hello VFS in Linux  
File Size: 18 bytes  
Inode Number: 893  
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical$ |
```

Practical 2: Mounting and Unmounting a File System

```
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical/Mount_Practical$ dd if=/dev/zero of=disk.img bs=1M count=20  
20+0 records in  
20+0 records out  
20971520 bytes (21 MB, 20 MiB) copied, 0.311566 s, 67.3 MB/s  
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical/Mount_Practical$ |
```

```
20971520 bytes (21 MB, 20 MiB) copied, 0.311566 s, 67.3 MB/s  
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical/Mount_Practical$ mkfs.ext4 -F disk.img  
mke2fs 1.47.0 (5-Feb-2023)  
Discarding device blocks: done  
Creating filesystem with 5120 4k blocks and 5120 inodes  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (1024 blocks): done  
Writing superblocks and filesystem accounting information: done  
saireddy@DESKTOP-HKQRL9G: ~/VFS_Practical/Mount_Practical$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ sudo apt update
[sudo] password for saireddy:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
94 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
e2fsprogs is already the newest version (1.47.0-2.4~explubuntu4.1).
e2fsprogs set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 94 not upgraded.
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ mkdir mnt
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ sudo mount -o loop disk.img mnt
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ df -h mnt
Filesystem      Size  Used Avail Use% Mounted on
/dev/loop0      15M   24K   14M   1% /home/saireddy/VFS_Practical/Mount_Practical/mnt
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ sudo touch mnt/sample.txt
ls -l mnt
total 16
drwx----- 2 root root 16384 Sep 25 11:01 lost+found
-rw-r--r-- 1 root root    0 Sep 25 11:03 sample.txt
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical$ |
```

Practical 3: Memory-Mapped Files using mmap()

```
GNU nano 7.2
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <string.h>

int main()
{
    int fd;
    char *ptr;

    fd = open("data.txt", O_RDWR | O_CREAT, 0644);

    if (fd < 0)
    {
        perror("open");
        return 1;
    }

    ftruncate(fd, 100);

    ptr = mmap(NULL, 100, PROT_READ | PROT_WRITE,
               MAP_SHARED, fd, 0);

    if (ptr == MAP_FAILED)
    {
        perror("mmap");
        close(fd);
        return 1;
    }

    strcpy(ptr, "Memory Mapped File in Linux");

    printf("Data: %s\n", ptr);

    msync(ptr, 100, MS_SYNC);

    munmap(ptr, 100);
    close(fd);

    return 0;
}
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical$ gcc mmap.c -o mmap
./mmap
Data: Memory Mapped File in Linux
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical$ |
```

```
Data: Memory Mapped File in Linux
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical$ cat data.txt
Memory Mapped File in Linuxsaireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical$ |
```

Practical 4: ext2/ext3/ext4 File System Information

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ dd if=/dev/zero of=ext.img bs=1M count=20
20+0 records in
20+0 records out
20971520 bytes (21 MB, 20 MiB) copied, 0.174223 s, 120 MB/s
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ mkfs.ext4 -F ext.img
mke2fs 1.47.0 (5-Feb-2023)
Discarding device blocks: done
Creating filesystem with 5120 4k blocks and 5120 inodes

Allocating group tables: done
Writing inode tables: done
Creating journal (1024 blocks): done
Writing superblocks and filesystem accounting information: done

saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ dumpe2fs -h ext.img
dumpe2fs 1.47.0 (5-Feb-2023)
Filesystem volume name: <none>
Last mounted on: <not available>
Filesystem UUID: 30526342-e460-4c07-93e6-5b207914e4a4
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index filetype extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags: signed_directory_hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 5120
Block count: 5120
Reserved block count: 256
Overhead clusters: 1350
Free blocks: 3764
Free inodes: 5109
First block: 0
Block size: 4096
Fragment size: 4096
Group descriptor size: 64
Reserved GDT blocks: 2
Blocks per group: 32768
Fragments per group: 32768
Inodes per group: 5120
Inode blocks per group: 320
Flex block group size: 16
Filesystem created: Fri Sep 25 11:07:04 2026
Last mount time: n/a
Last write time: Fri Sep 25 11:07:04 2026
Mount count: 0
Maximum mount count: -1
Last checked: Fri Sep 25 11:07:04 2026
Check interval: 0 (<none>)
Lifetime writes: 21 kB
Reserved blocks uid: 0 (user root)
Reserved blocks gid: 0 (group root)
First inode: 11
Inode size: 256
Required extra isize: 32
Desired extra isize: 32
Journal inode: 0
Default directory hash: half_md4
Directory Hash Seed: 7d6892bb-1681-4e6d-95ae-c0bbc6066875
Journal backup: inode blocks
Checksum type: crc32c
Checksum: 0x9f99f06f
Journal features: (none)
Total journal size: 4096kB
Total journal blocks: 1024
Max transaction length: 1024
Fast commit length: 0
Journal sequence: 0x00000001
Journal start: 0

saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ |
```

```
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ dumpe2fs -h ext.img 2>/dev/null | grep -E "Filesystem features|Block count|Block size|Inode count"
Filesystem features: has_journal ext_attr resize_inode dir_index filetype extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Inode count: 5120
Block count: 5120
Block size: 4096
saireddy@DESKTOP-HKQRL9G:~/VFS_Practical/Mount_Practical/Mmap_Practical/Ext_Practical$ |
```